IMEN574 / IDSC574 · PROGRAMMING FOR AX

10

Trees and Ensembles

Sangwoo Mo · POSTECH
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Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Trees and Ensembles

01 **Partitioning**

Fit local rules

02 **Averaging**

Reduce unstable variation

03 **Sequential Correction**

Improve the current fit

01 · Partitioning

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Tree Model Families

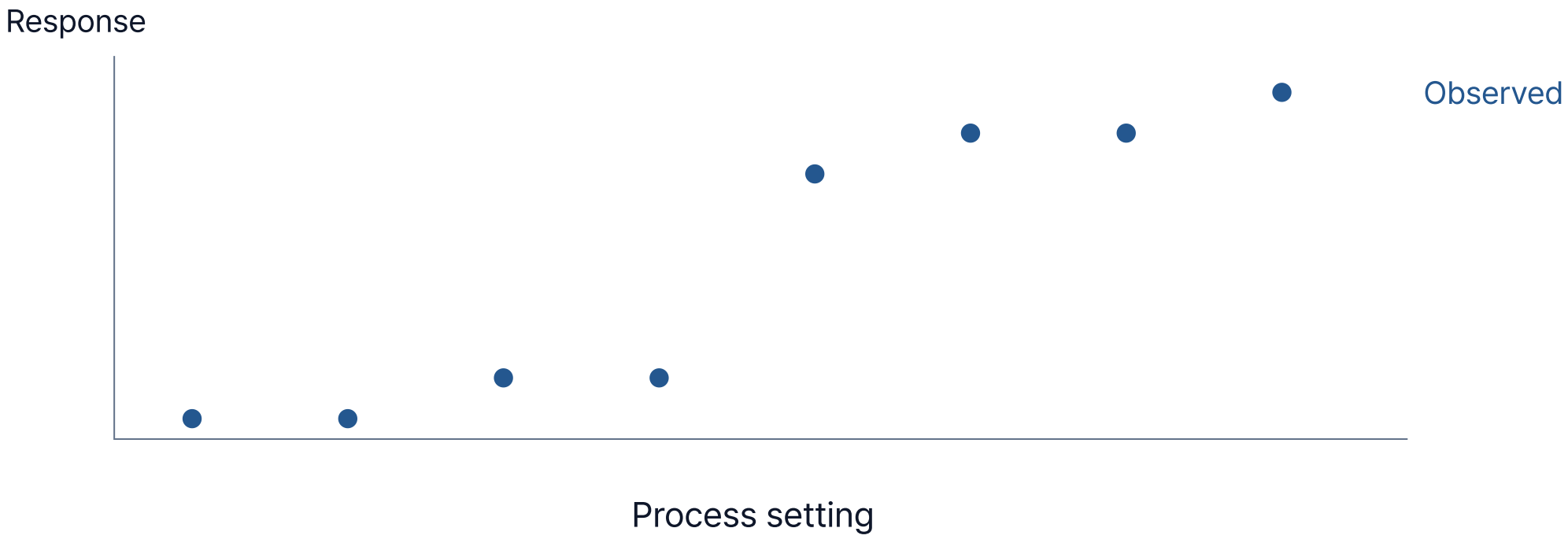
One split mechanism supports different target types

Target	Leaf Output	Split Criterion
Continuous	Mean response	Squared error
Categorical	Class proportions	Gini or entropy

The leaf output changes with the prediction target

Why Local Rules

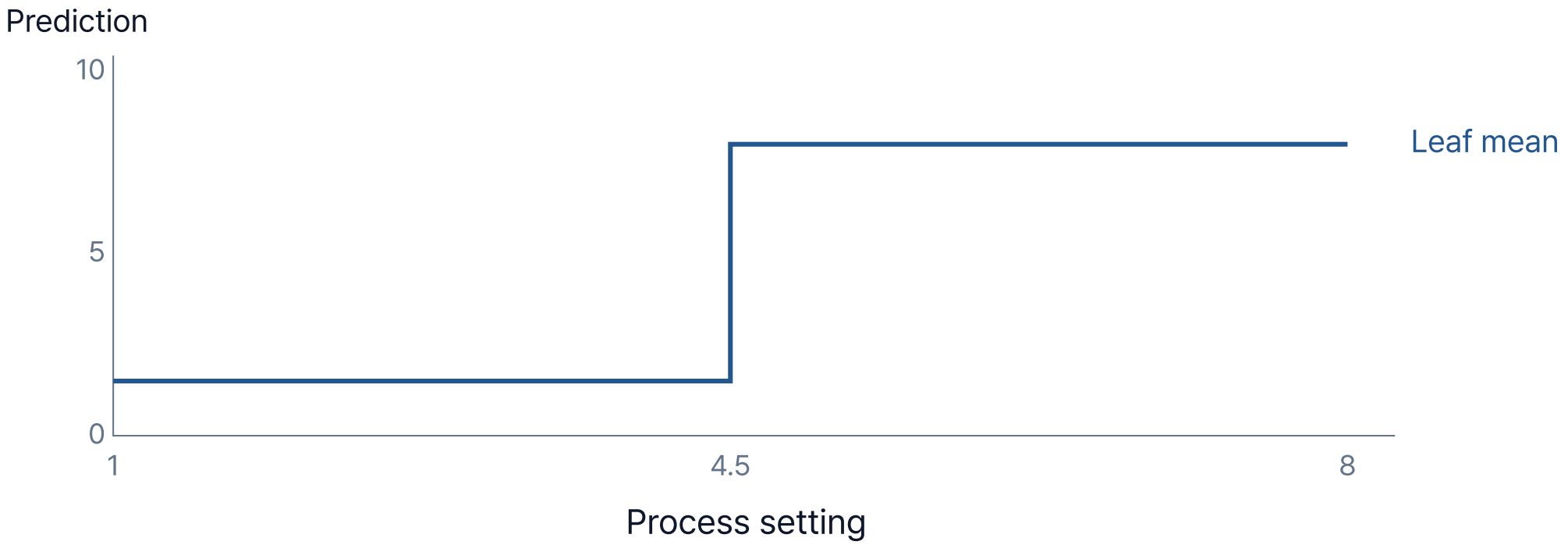
One slope cannot match a step change



The response jumps between settings four and five

A Regression Stump

One threshold creates two constant predictions



Each side predicts its own training mean

Split Objective

Choose the cut that reduces within-leaf error



Sum of squared residuals

The cut removes most between-group variation

Searching Candidate Cuts

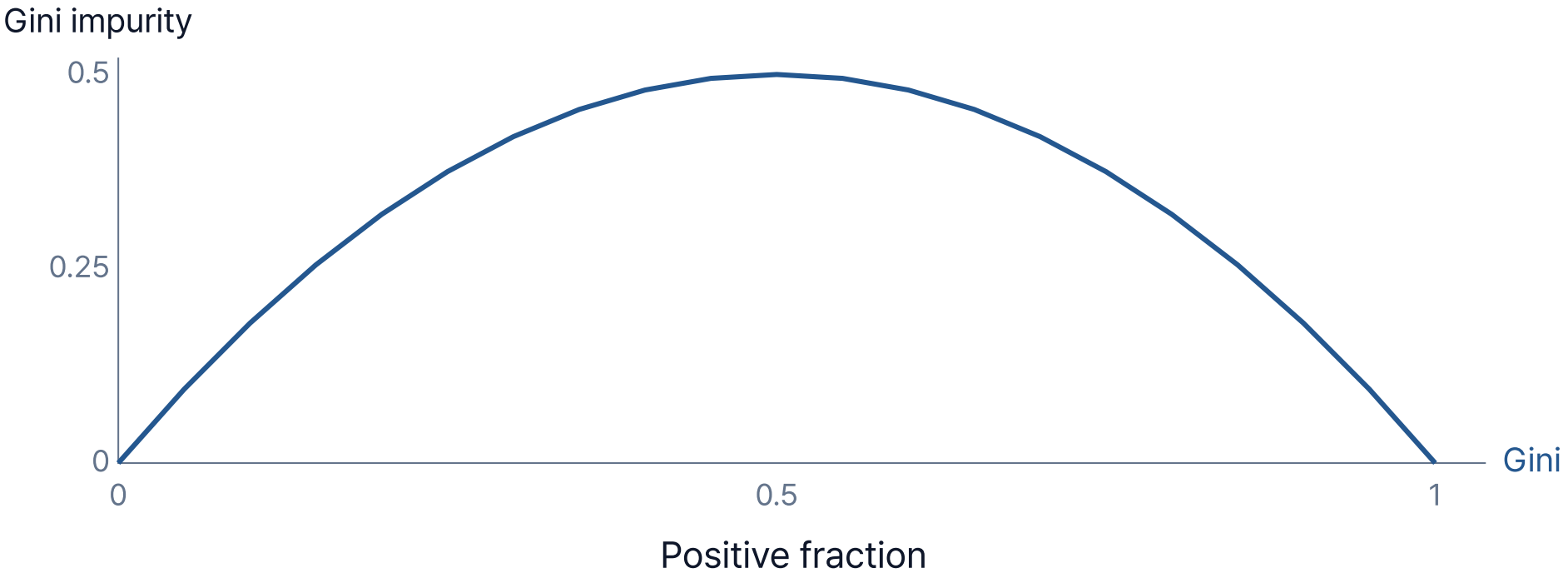
Evaluate boundaries between sorted feature values

```
x=np.arange(1,9,dtype=float)
y=np.array([1,1,2,2,7,8,8,9.])
for t in [2.5,4.5,6.5]:
    a,b=y[x<=t], y[x>t]
    sse=((a-a.mean())**2).sum()+((b-
b.mean())**2).sum()
    print(t, round(sse,2))
```

```
2.5 50.0
4.5 3.0
6.5 50.0
```

Classification Impurity

A mixed leaf has more uncertainty than a pure leaf

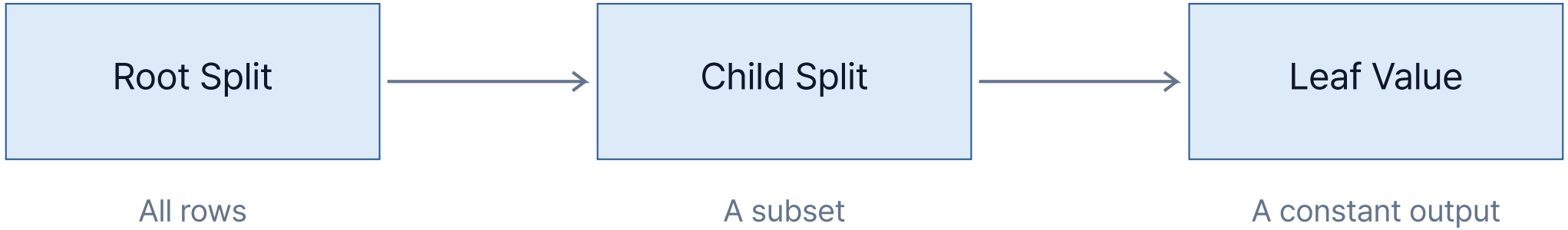


$$\text{Gini} = 1 - \sum_k p_k^2$$

A balanced binary leaf has the largest impurity

Recursive Partitioning

Later questions refine the remaining region



Every path restricts the rows reaching one leaf

Interaction Without Products

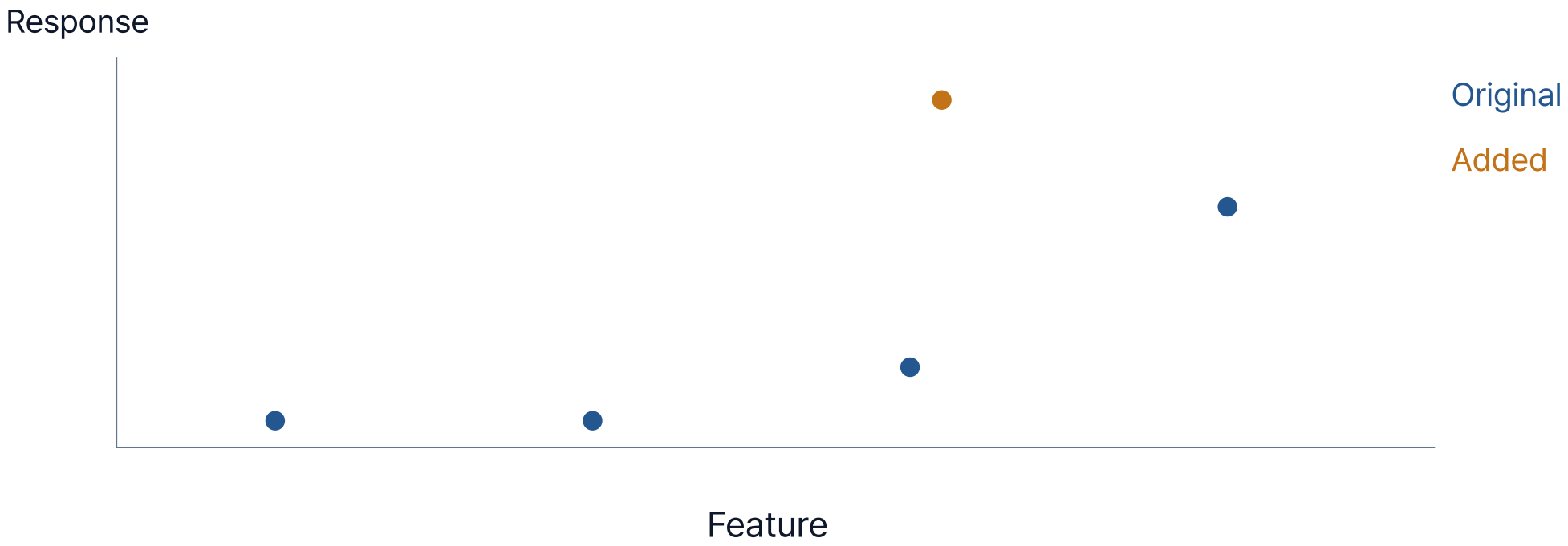
A later cut can depend on an earlier condition

	$x_2 < 0$	$x_2 \geq 0$
$x_1 < 0$	Low	Low
$x_1 \geq 0$	Low	High

The positive response occupies one feature quadrant

Caveat: Unstable Splits

Small data changes can change a greedy decision



One added observation changes the local tradeoff

Depth and Leaf Size

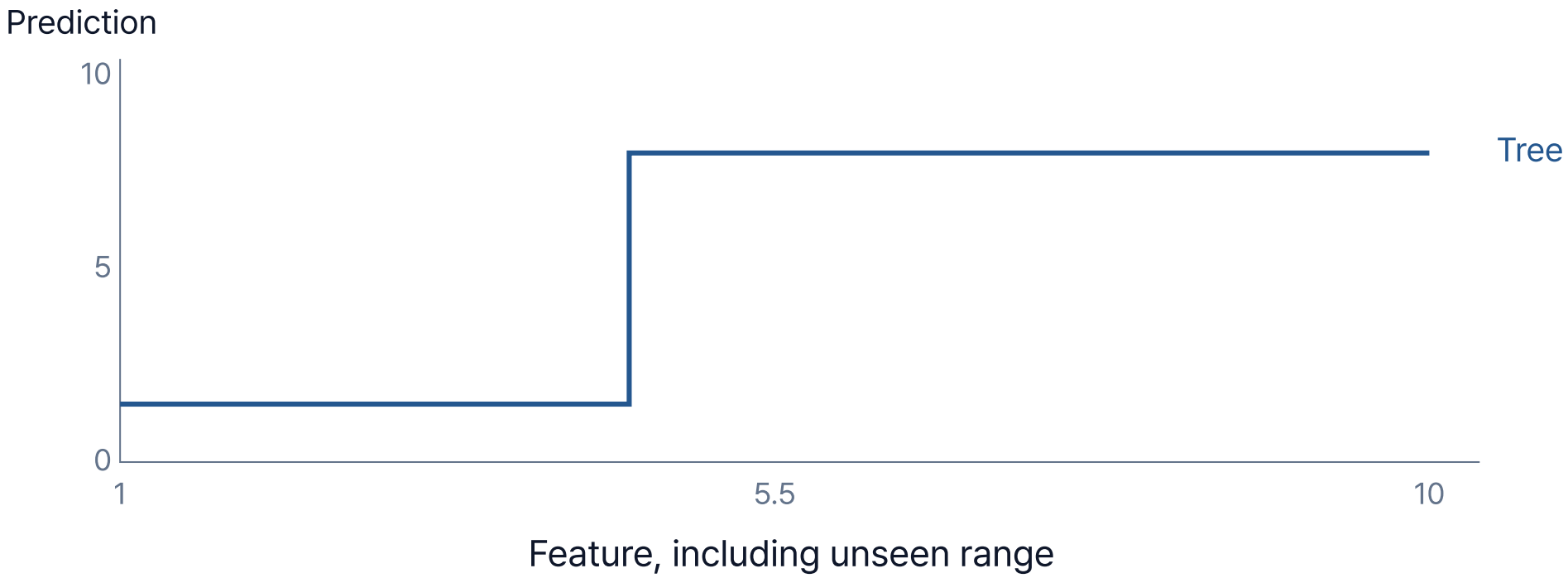
Limit how narrowly the model memorizes training rows

Depth Budget	Possible Rows/Leaf
0	8
1	4
2	2
3	1

Smaller leaves use less evidence for each prediction

Extrapolation Limit

A tree keeps leaf predictions outside the observed range



The final leaf remains constant beyond the training maximum

02 · Averaging

01 **Partitioning**

Fit local rules

02 **Averaging**

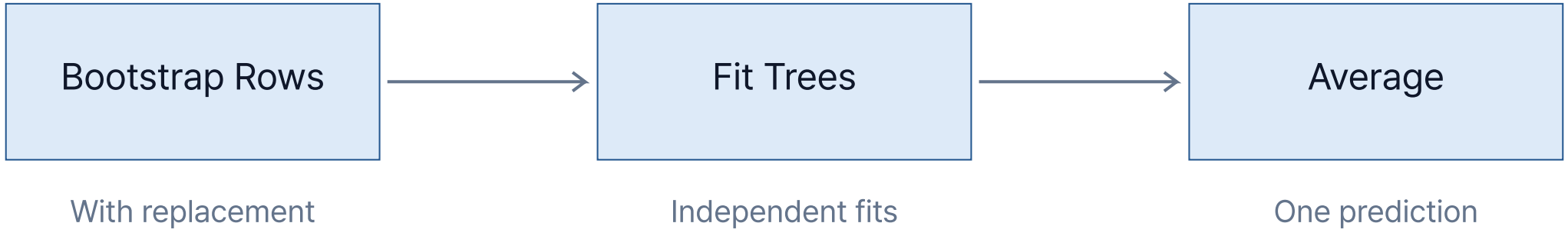
Reduce unstable variation

03 **Sequential Correction**

Improve the current fit

Bagging Estimators

Fit varied datasets and average their predictions



Averaging combines trees fitted to different row samples

Bootstrap Sampling

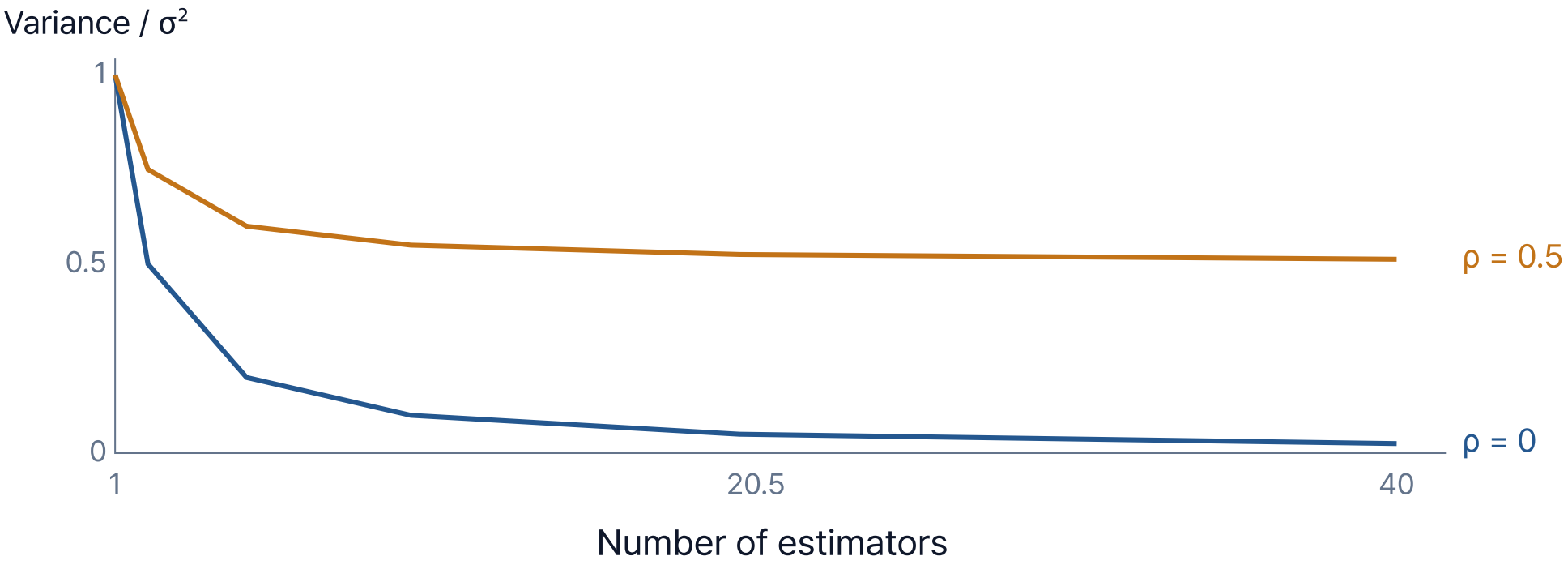
Repeated rows and omitted rows appear in each sample

	Draw 1	Draw 2	Draw 3	Draw 4	Draw 5
Original	1	2	3	4	5
Sample A	2	2	3	5	5
Sample B	1	1	2	4	5

Each bootstrap sample has the same size with replacement

Variance of an Average

Correlation limits the benefit of more estimators



$$\mathrm{Var}(\mathrm{mean}) = \sigma^2[\rho + (1 - \rho)/M]$$

Correlated errors leave a variance floor

Random Feature Subsets

Decorrelate trees by restricting candidate features

	Candidate 1	Candidate 2
Tree A	x_1	x_3
Tree B	x_2	x_3
Tree C	x_1	x_2

Different candidate sets encourage different split choices

Out-of-Bag Predictions

Predict each training row using trees that omitted it

	Tree 1	Tree 2	Tree 3
Row A	In	Out	Out
Row B	Out	In	Out
Row C	In	Out	In

A row can be evaluated by trees that did not fit it

Regression Averaging

Average numeric predictions from the trees

	Tree A	Tree B	Tree C	Mean
One Input	6	8	10	8

Three leaf values produce a mean prediction of eight

Classification Averaging

Combine class scores before selecting an action

	Tree A	Tree B	Tree C	Mean
Positive Probability	0.2	0.6	0.7	0.5

The mean score and threshold determine the final label

Feature Importance Bias

A flexible split variable can look important by chance

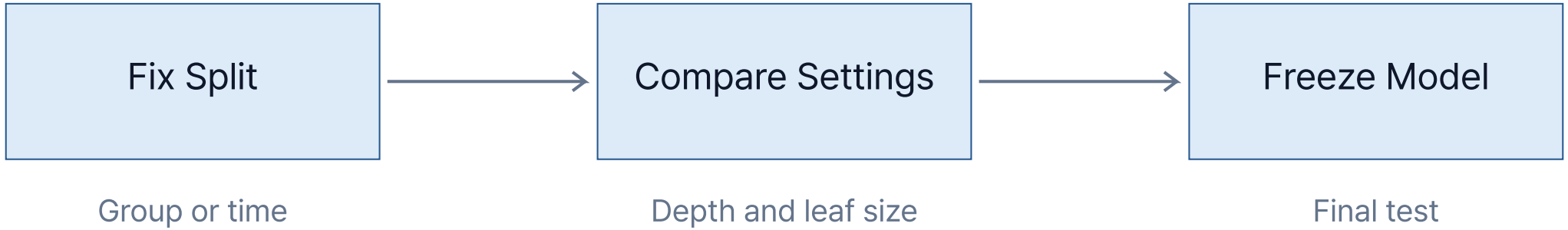


Illustrative impurity importance

More candidate cut points can inflate training-based importance

Forest Validation

Tune structure with the same split protocol



More trees do not repair a leaking split

Forest Resource Cost

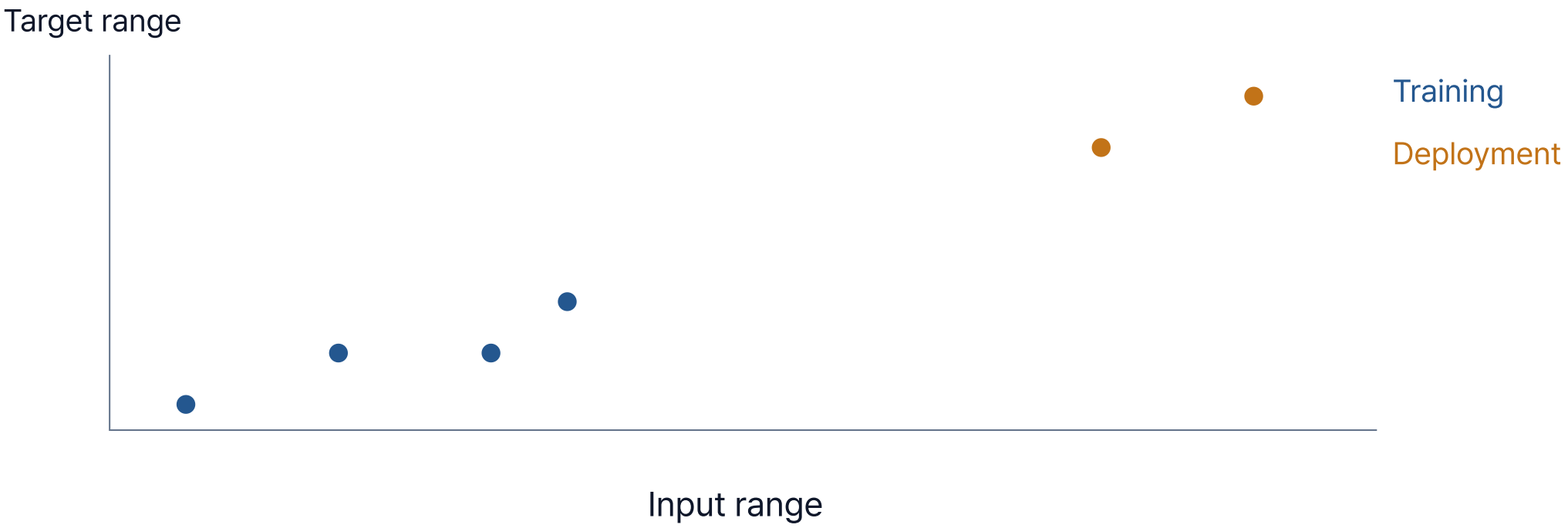
Prediction cost grows with trees and traversal depth

Trees	Depth	Path Steps
100	8	800
500	8	4000
500	16	8000

The rough traversal budget grows with both factors

Forest Failure Case

Shared missing coverage survives averaging



All trees can share the same uncovered deployment region

03 · Sequential Correction

01 **Partitioning**

Fit local rules

02 **Averaging**

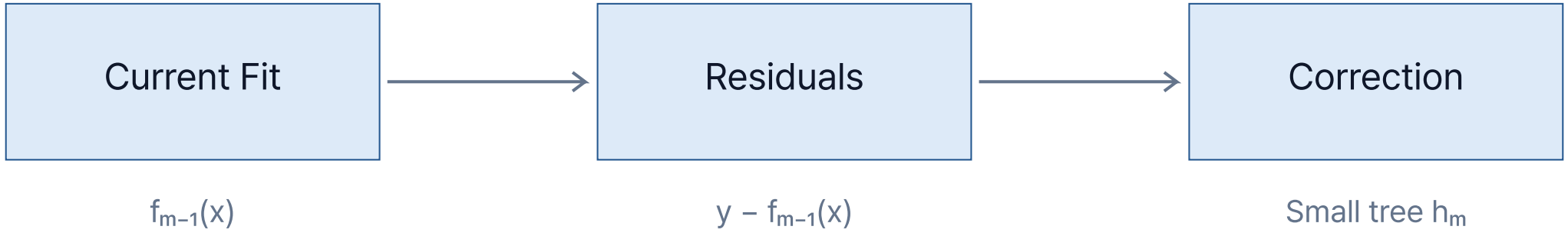
Reduce unstable variation

03 **Sequential Correction**

Improve the current fit

Gradient Boosting

Each new model corrects the current ensemble



For squared loss the correction target is the residual

First Boosting Residuals

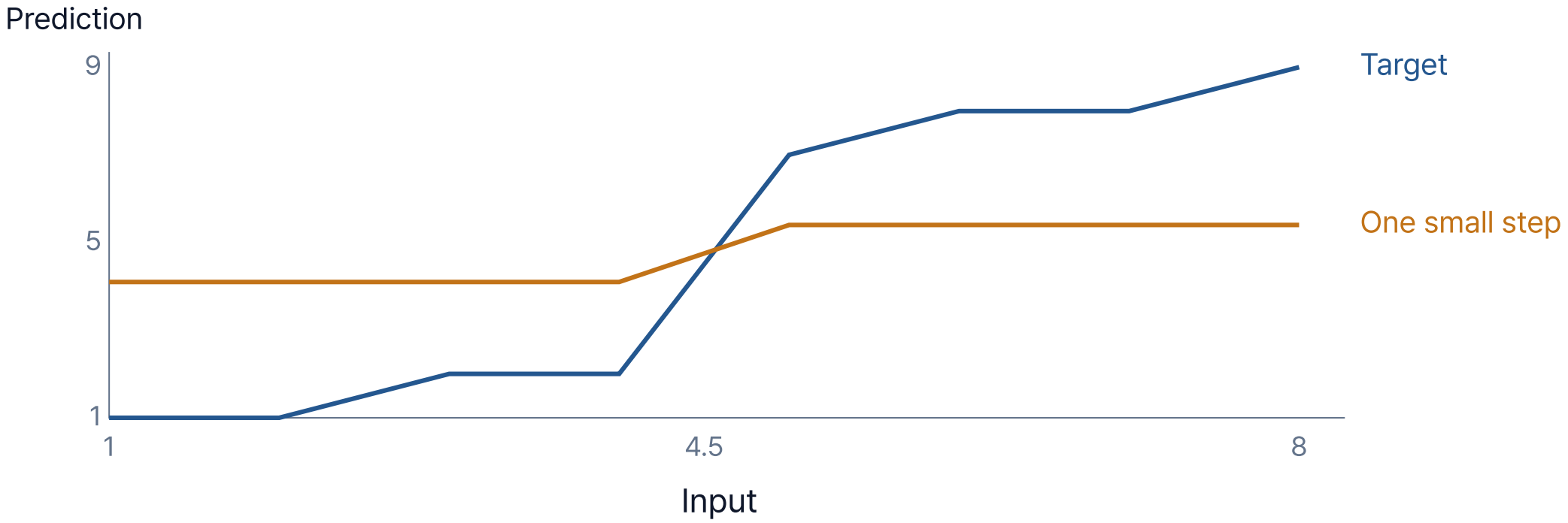
Start from a constant fit and inspect what remains

	1	2	3	4	5	6	7	8
Target	1	1	2	2	7	8	8	9
Initial Mean	4.75	4.75	4.75	4.75	4.75	4.75	4.75	4.75
Residual	-3.75	-3.75	-2.75	-2.75	2.25	3.25	3.25	4.25

Residuals are negative on the left and positive on the right

Shrinkage Step

A learning rate controls the size of each correction



$$f_m = f_{m-1} + \eta h_m$$

A small correction moves toward the target without replacing the fit

Boosting Loss Gradients

Other losses use negative gradients as pseudo-targets

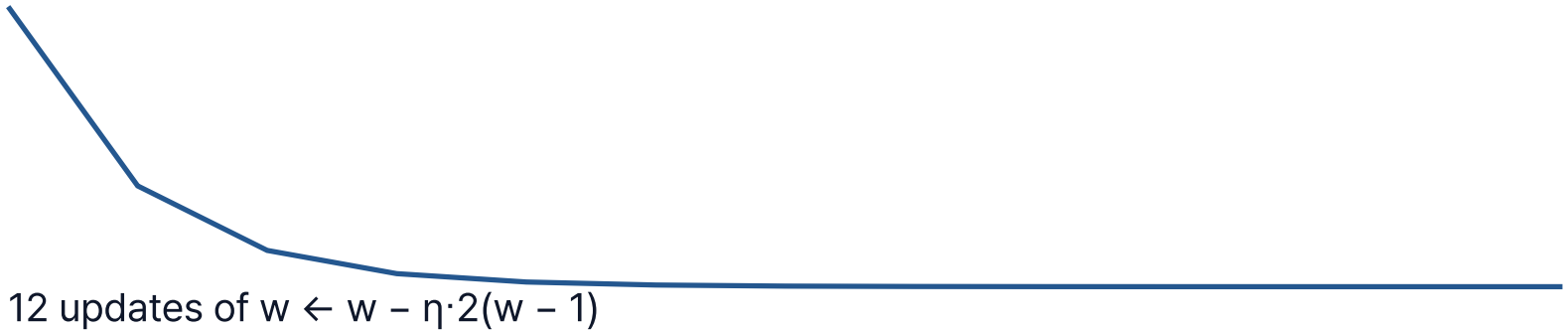
Loss	Correction Signal
Squared error	y − f
Binary log loss	y − p

The residual idea extends through the loss gradient

Demo: Learning Rate

Increase the learning rate until the updates stop settling

Learning rate 0.2

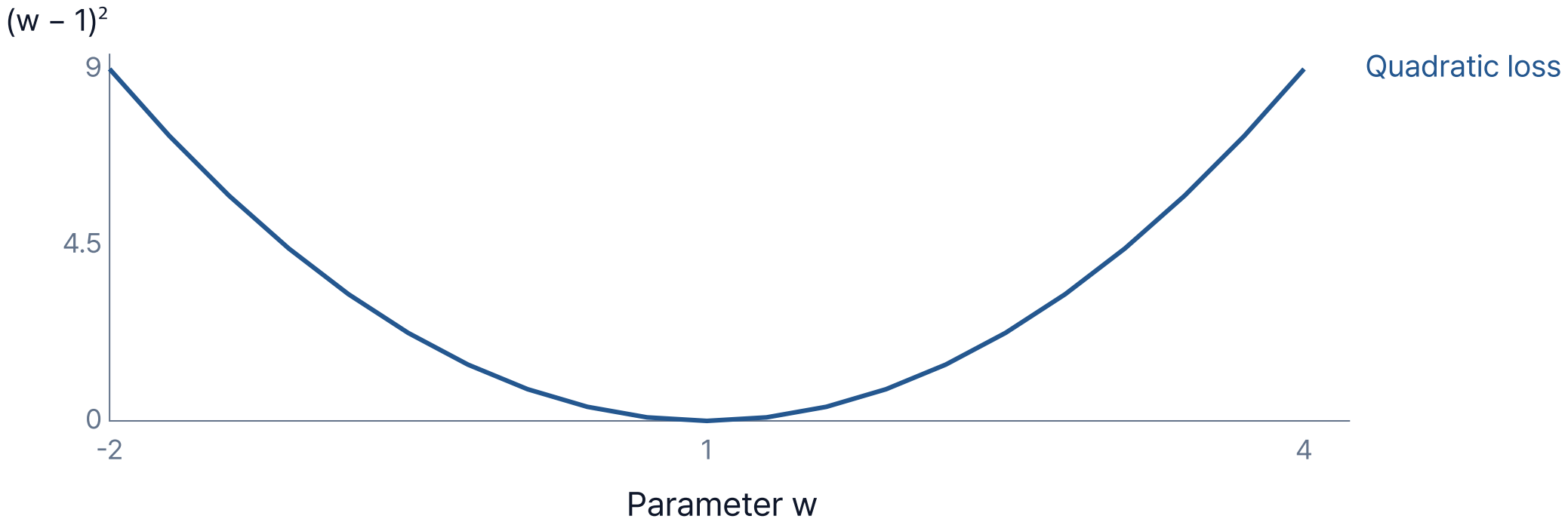


Learning rate 0.20 | w after 12 steps 1.00653 | loss 0.00004265

For this quadratic, $0 < \eta < 1$ converges; $\eta \geq 1$ does not converge from $w = 4$

Answer: Correction Size

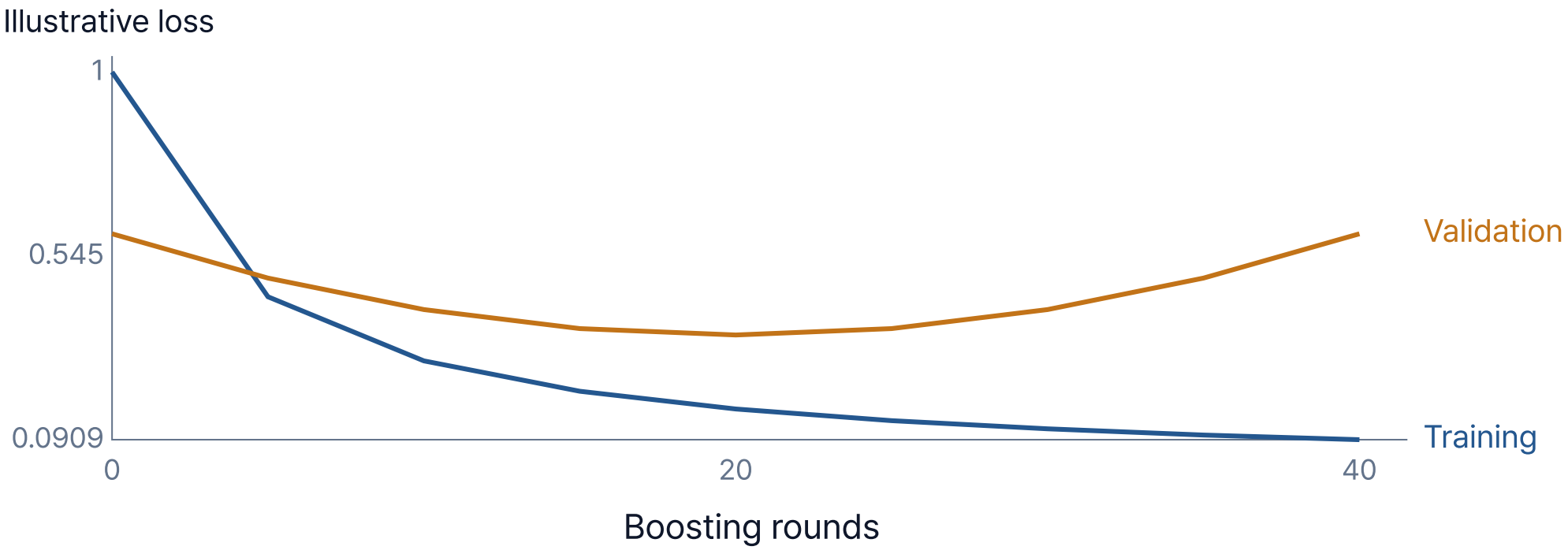
A useful direction can fail with a large step



Crossing the minimum repeatedly can increase the loss

Early Stopping for Boosting

Select the round using validation rather than training loss



The validation minimum occurs before the final round

Modern Boosting Libraries

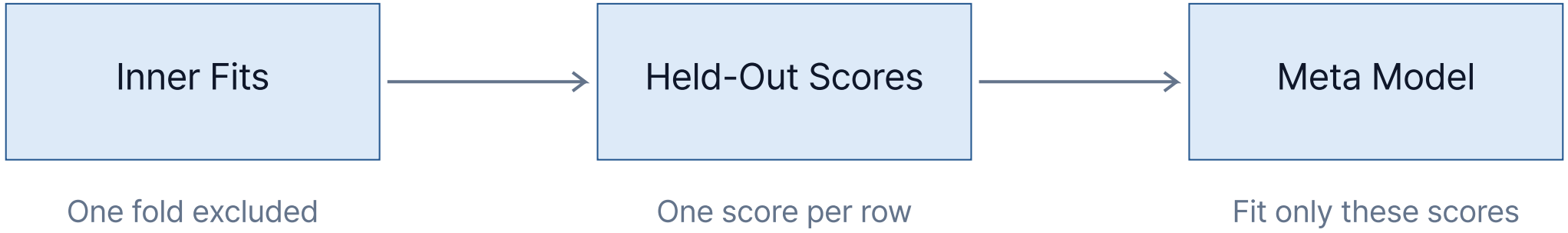
Implementation choices change speed and constraints

Library	Useful Idea
XGBoost	Regularized tree boosting
LightGBM	Histogram-based split search
HistGradientBoosting	Histogram boosting in sklearn

Compare implementations under one dataset and evaluation protocol

Stacking Without Leakage

Train the combiner on out-of-fold predictions



Training-fit predictions give the combiner an unfair view

A Practical Model Comparison

Keep a simple reference and change one family at a time

Family	Strength	Check
Linear	Fast reference	Missed interactions
Forest	Stable nonlinear fit	Latency and range
Boosting	Flexible corrections	Tuning and leakage

A model family is a candidate rather than a guaranteed winner

Takeaway: Trees and Ensembles

01 Tree leaves are local constant estimates

02 Averaging benefits from varied errors

03 Boosting corrects the current loss signal